

Speaker & Audio DAC Fundamentals

A complete, crystal-clear guide to digital audio playback, Delta-Sigma DAC reconstruction, Dynamic Range Compression (DRC), anti-pop ramping, and speaker amplification using the ES8311 and ESP32-S3.

Target CODEC: **Everest Semi ES8311 (QFN-20)** Host Controller: **ESP32-S3 (I2S DMA + I2C)** Audio Quality: **24-Bit / 110 dB SNR**

1. Core Fundamentals: How Digital Audio Becomes Sound

Digital sound stored in memory is a stream of numeric PCM samples. The **Speaker & DAC subsystem** reconstructs these numbers into continuous analog electrical waves that push a physical speaker cone back and forth to create sound in the air.

🔊 Loudspeaker & Power Amp Driving (Differential BTL)

Loudspeakers require significant electrical current (e.g. 1W to 3W). The ES8311 acts as a high-fidelity pre-driver, outputting clean differential signals (`OUTP` and `OUTN`) to an external Class-D power amplifier (such as NS4168, MAX98357, or NS4150).

Advantage: Differential output doubles voltage swing (+6 dB) and cancels WiFi RF noise and power supply hum.

🎧 Direct Headphone / Earphone Driving (Single-Ended)

The ES8311 contains an on-chip, low-impedance headphone driver. By setting `HP_SW = 1` in Register `0x13`, it can directly drive 16Ω to 32Ω earphones through DC-blocking capacitors without requiring an external amplifier.

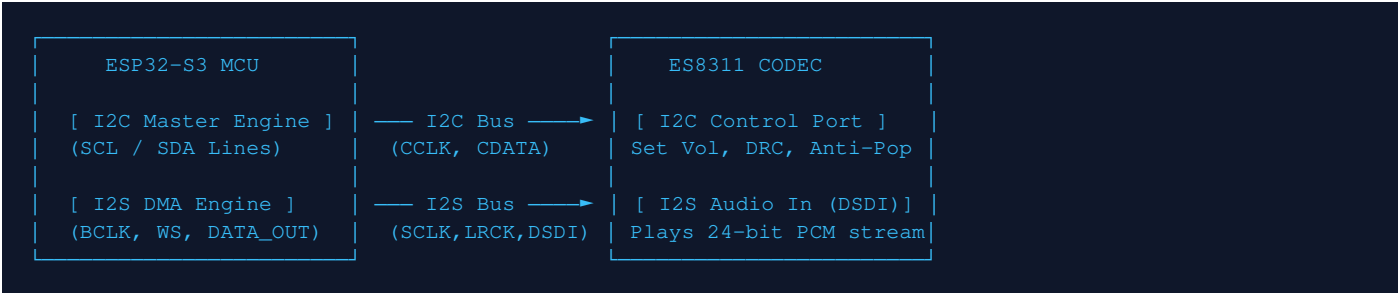
2. Why Do We Need the ES8311 Audio DAC?

Driving speakers directly from microcontroller PWM or internal 8-bit DACs produces thin, noisy, distorted audio. The **ES8311 Audio CODEC** provides studio-grade playback in hardware:

Playback Feature	Microcontroller Built-In DAC / PWM	ES8311 Dedicated Audio DAC
Resolution	8-Bit / 10-Bit (High quantization noise)	24-Bit Multi-Bit Delta-Sigma (Studio Quality)
Signal-to-Noise (SNR)	~50 dB to 60 dB (Loud hiss & whine)	110 dB SNR (Ultra-quiet black background)
THD+N Distortion	~1% to 5% (Distorted bass & highs)	-80 dB / 0.01% THD+N (Pristine clarity)
Anti-Pop & Click	Loud pop on speaker when boot/shutdown	Hardware Soft Volume Slew-Rate Ramping
Dynamic Control	Must be coded in complex DSP software	Hardware Dynamic Range Compression (DRC)
CPU Overhead	High CPU load during timer interrupts	Zero (Streams via I2S DMA automatically)

3. Dual-Plane System Architecture: I2C Control vs I2S Playback

The ESP32-S3 communicates with the ES8311 using two dedicated, concurrent interfaces:



- **Control Plane (I2C):** Low-speed configuration bus (100 kHz - 400 kHz). Adjusts volume, enables DRC compression, and configures anti-pop ramps.
- **Data Plane (I2S):** High-speed audio streaming bus. Pushes PCM audio frames continuously from ESP32 flash memory or network streams into the DAC with zero CPU latency.

4. The 7-Stage Playback Signal Processing Pipeline

When digital audio enters the ES8311, it passes through 7 hardware processing stages:



5. Pinout & Pin Role Directory for Playback (QFN-20)

Pin #	Pin Name	Type	Playback Hardware Description & Connection Guide
1	CCLK	Input	I2C Clock: Connect to ESP32 I2C SCL with a 3.3 kΩ pull-up resistor.
2	MCLK	Input	Master Clock: Sourced from ESP32 ($256 \times F_s = 12.288 \text{ MHz @ } 48 \text{ kHz}$).
3	PVDD	Power	I/O Pad Power Rail: Connect to 3.3V with 0.1 μF bypass cap.
4	DVDD	Power	Digital Core Power Rail: Connect to 1.8V or 3.3V with 0.1 μF cap.
5	DGND	Ground	Digital Ground: Tie to main system ground plane.
6	SCLK	Input	I2S Bit Clock: Synchronizes incoming PCM bits on DSDIN.
8	LRCK	Input	Frame Clock / Word Select (WS): Left/Right channel frame toggle ($= F_s = 48 \text{ kHz}$).
9	DSDIN	Input	DAC Serial Audio Data In: Connect to ESP32 I2S Data Out pin.
10	AGND	Ground	Analog Ground: Quiet ground reference for analog DAC and output buffers.
11	AVDD	Power	Analog Power Rail: 3.3V clean analog supply with 1 μF 0.1 μF bypass caps.
12	OUTP	Output	Positive Analog Audio Output: Connect to IN+ of power amp or Left earphone.
13	OUTN	Output	Negative Analog Audio Output: Connect to IN- of power amp (Differential 180° phase).
14	DACVREF	Analog Ref	DAC Voltage Ref: Connect 1 μF bypass capacitor to AGND. Essential for 110 dB SNR!
16	VMID	Analog Ref	Mid-Rail DC Bias (AVDD/2): Connect 1 μF cap to AGND. Prevents startup pop noise.
19	CDATA	I/O	I2C Data Line: Connect to ESP32 I2C SDA with 3.3 kΩ pull-up resistor.
20	CE	Input	I2C Address Select: Tie to GND for address 0x18; tie to 3.3V for address 0x19.
21 (EP)	Exposed Pad	Ground	Thermal & Ground Pad: Solder bottom center pad directly to PCB Ground plane.

6. The 3 Speaker & Output Wiring Schematics

Scheme 1: Differential Speaker Output with Power Amp (Loudspeakers)

OUTP (Pin 12) —||— [1uF Cap] —> Power Amp IN+ | OUT+ —> Speaker (+)

OUTN (Pin 13) —||— [1uF Cap] —> Power Amp IN- | OUT- —> Speaker (-) [4Ω / 8Ω]

* Register: Reg 0x13 = 0x00 (Line Out Mode)

Scheme 2: Direct Headphone / Earphone Drive (16Ω - 32Ω)

OUTP (Pin 12) —||— [100uF - 220uF] —> 3.5mm Tip (Left/Mono)

OUTN (Pin 13) —||— [100uF - 220uF] —> 3.5mm Ring (Right) | Sleeve —> AGND

* Register: Reg 0x13 = 0x10 (Enable Headphone High-Current Driver Mode)

Scheme 3: Single-Ended Line Out (Aux Out to PC / Mixer)

OUTP (Pin 12) —||— [1uF Cap] —> Aux Signal Line Out | AGND —> Shield / Ground

* Pin 13 (OUTN) is left unconnected (NC).

7. 5-Step ESP32-S3 Playback Bring-Up Recipe

To start playing 24-bit audio at 48 kHz, send this exact sequence over I2C during system boot:

```
// 1. Soft Reset & Start Reference Voltage Pre-charge
i2c_write(0xFA, 0x01); // Soft reset all registers to default
delay_ms(10);
i2c_write(0x0D, 0x01); // Start VMID & DACVREF fast pre-charge
delay_ms(25); // Wait for capacitors to stabilize (prevents pop click)
i2c_write(0x0D, 0x02); // Switch VMID to normal low-noise operating state

// 2. Configure System Clocks (Slave Mode, MCLK from ESP32)
i2c_write(0x00, 0x80); // Enable State Machine in slave mode
i2c_write(0x01, 0x35); // Turn ON MCLK, BCLK, DAC digital clock, and DAC analog clock
i2c_write(0x04, 0x10); // DAC Oversampling Ratio = 64x (or 0x40 for 256x)

// 3. Configure Serial Data Port Input (I2S Format, 24-bit depth)
i2c_write(0x09, 0x00); // SDP_IN: Play Left channel, 24-bit word length, I2S, unmuted

// 4. Configure Anti-Pop Soft Ramping & Master Volume
i2c_write(0x37, 0x88); // 0.25 dB / 512 LRCK slew rate, DACEQ bypassed
i2c_write(0x32, 0xBF); // Master Volume = 0 dB Unity Gain (100% Volume)

// 5. Power Up Analog DAC Output Stage
i2c_write(0x13, 0x00); // HPSW = 0 (Line Out mode for external speaker amp)
i2c_write(0x12, 0x00); // Power ON DAC analog core and enable reference output
```

8. Speaker Troubleshooting & Cheat Sheet

Symptom	Root Cause	Hardware / Software Solution
Loud Pop on Startup / Shutdown	Abrupt power toggle without pre-charge	Allow VMID 25 ms to charge before enabling DAC (Reg 0x12). Enable soft ramping in Reg 0x37 = 0x88.
Complete Silence	Volume default is mute (0x00)	Register 0x32 defaults to -95.5 dB. You must write 0xBF (0 dB) to enable sound. Verify Reg 0x12 = 0x00.
Distorted / Raspy at Loud Volume	Input audio clipping speaker amp	Enable DRC Compressor: write 0x80 to Reg 0x34 and 0xF0 to Reg 0x35 (-6 dB peak limit).
Direct Mic-to-Speaker Sidetone	Need zero-latency local mic monitoring	Set Reg 0x44[7] = 1 (ADC2DAC_SEL = 1) to stream microphone directly into speaker in hardware.